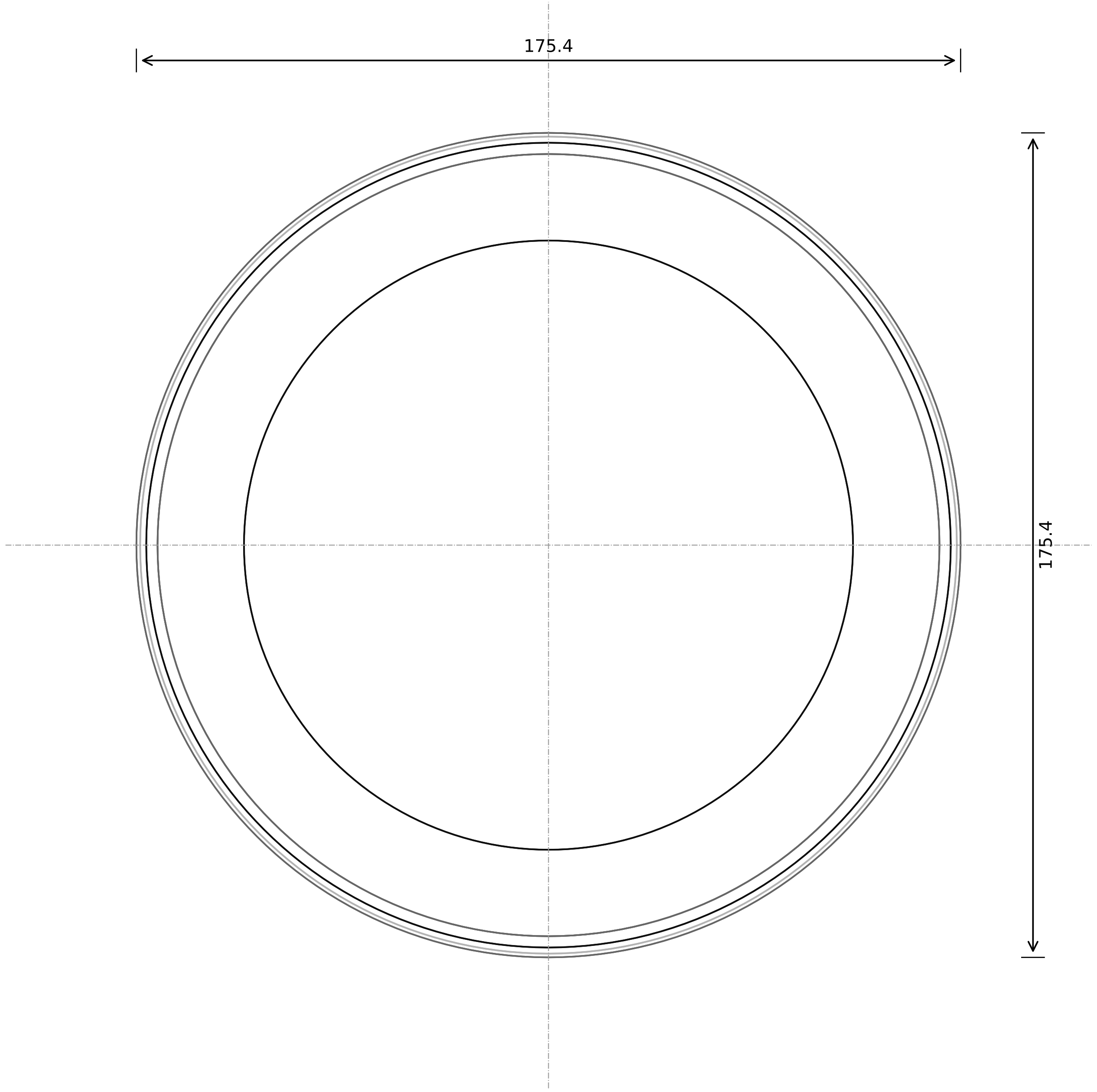
the 60 — v15 — sheet 0: the ASSUMED numbers (every one of these must be verified on the real part before the drawing is trusted)

2026-09-06. Stack: plate 8.0, adapter 14.78-16.38 (socket to 23.68), code ring under the crown at 31.53, seat 26.0, glass 26.50-28.48, lens to 30.98, knob top 34.38, overall 35.88 with the pad. Ø175.

1. cover lens: Ø140.0 x 2.5	specified separately; carrier rim and lip overlap designed to it
2. bonding tape: 0.5	double-sided foam tape between the seat and the glass edge (RULING 5 Sep: glue allowed for the display)
3. panel back component area: 60.0 x 68.36 x 1.5 at -14.0	position from the drawing's back view
4. Active Cooler position: from Pi x 2.5, fan from x 43.0	no CAD body; footprint and fan position from photographs
5. cooler base above the board: 2.8	SoC 2.4 + thermal pad
6. fan intake plenum: 8.0	minimum free height above the blower
7. micro-HDMI receptacles: x 26.0/39.5	not in the vendor STEP; drawn as envelopes
8. HDMI ribbon plugs: 12x8x5 micro end, 21x12x6 A end	flat FPC HDMI cable, 200 mm
9. adapter board thickness and part heights: 1.6; parts 2.5, underside 1.0, flex connector 2.0	outline, holes, socket size and position PUBLISHED (datasheet drawing)
10. display connect board: 30.0 x 46.0 x 1.2, connectors 2.0	part of the adapter kit; size scaled from its photograph, hole positions assumed at the corners
11. Pi standoffs and washers: M2.5 x 12.0 hex, 0.5 washers	the adapter stack: closing-plate land / washer / Pi / standoff / adapter
12. USB-C up-angle plug: 12 x 12.5 x 6.5 head	measure the lead
13. audio / motion / touch / converter boards: envelopes	electrical layout not done
14. barrel jack body: 14.4 x 9.0 x 11.0 on a vertical board	PJ-063AH; CAD body to fetch
15. right-angle barrel plug: Ø8.5 body	measure the brick's plug
16. MOTOR_BASE_OD / MOTOR_BASE_H: 30.0 / 4.3	carried from v9
17. MOTOR_BAND_T: 0.6	carried from v9
18. servo pushrod: 3 mm bridge to the tab	carried from v9; servo on the plate in a frame
19. LED strip: 5.0 wide x 1.6 thick	any 5 mm addressable strip; a wider strip raises the halo band
20. blower inlet and outlet: inlet Ø24.0 in the face, outlet 21.0 x 7.0 from 1.5 up	from the datasheet drawing (outline, holes and leads PUBLISHED)
21. blower gaskets: 0.3 closed-cell foam	under the inlet face and the hood
22. closing-plate gasket: 0.2 die-cut	in a groove of the recess floor, on every sealing wall
23. Pi 5 fan connector current rating: 0.13 A drawn (0.20 max)	the connector sits in the vendor STEP; its rating is not published
24. encoder shim family: [0.7, 0.8, 0.9, 1.0, 1.1]	printed; the height is chosen on the bench

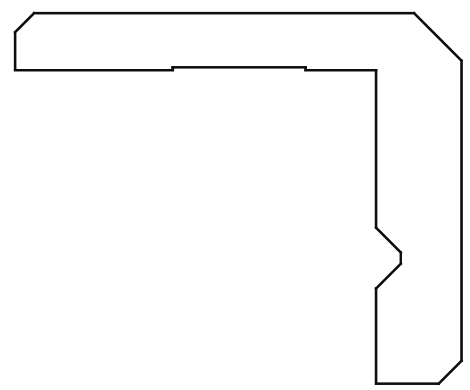
Build-time changes against V11-SPECIFICATION.md are listed in V11.md. Sheets 1-11 follow: one per made part.

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 58 to 90, enlarged)



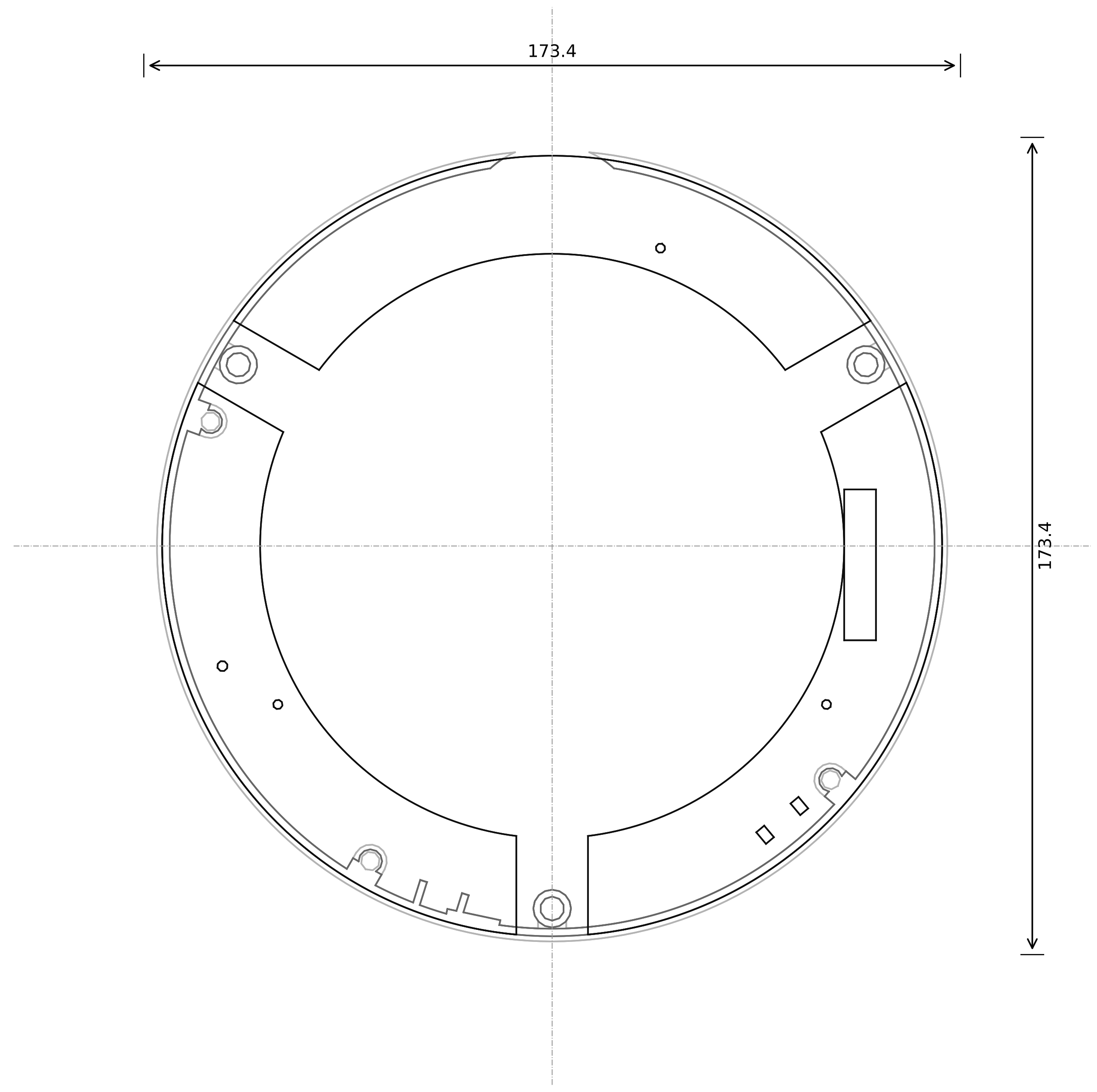
SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

1. outside Ø175.4, bore Ø166.4 (0.45 to the panel's chin corners); wall 4.5
2. top rim: lip inner edge Ø128.4, 1.0 inner chamfer, 20 flat, 2.5 outer chamfer (RULING 6 Sep: rim >= 20); lip 3.0 thick, top at z 34.38
3. outer chamfer 2.5 x 45° smooth; inner chamfer 1.0; skirt chamfer 1.2
4. V-groove 90°, root r 84.50399999999999, z 19.9-23.1
5. code RING recess 0.15 deep in the crown's underside, r 72.5-79.5 (v13: the encoder reads the crown, not the bore - the bore is plain apart from the V-groove)
6. diamond knurl 78 starts x 2, 1.0 deep, helix 36.8°, 3 whole rows z 16.9-31.1; skirt bottom z 14.9
7. wall under the knurl root 3.5

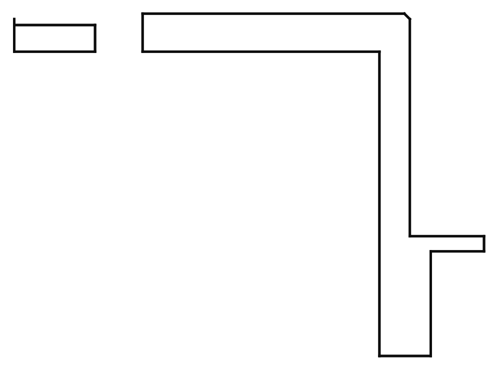
the 60 — v15	**Knob**
Cadrane · sheet 1 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: PETG (prototype) / CNC 6082-T6, hard anodised (production)	**Part: knob_body**
Process: print top face down, 0.16 mm; no supports	Overall 175.4 x 175.4 x 19.5
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,	Volume 69.36 cm³
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 57 to 89, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

1. wall r 81.204-82.80399999999999, z 8.0-26.0
2. seat flange r 62.0-82.80399999999999, z 24.0-26.0: its top face is FLAT (nothing rises above it) with 0.3 chamfers on both edges; three Ø2.0 H7 through-holes (ream) at [70, 210, 330]°, r 67.25 for the display's dowel pins, each in a Ø5.0 boss 4.0 under the flange
3. LED-strip band r 82.80399999999999-83.90399999999998, z 8.0-13.5 (strip r 84.00399999999998-85.60399999999997, 53 LEDs, 2.54 A full white); diffuser lip r 82.80399999999999-86.704, z 13.5-14.3
4. wheel posts Ø8.0 at r 76.93, az [30, 150, 270], to z 18.9; bush bores Ø5.1; wall windows 15.2 wide, open to the top from z 18.6; the seat flange is cut away above each wheel
5. pillars Ø7 at r 77.204, az [160, 240, 320], to z 16.0; M3 inserts from below (the plate screws) - nothing screws in from above
6. encoder at 310.0°, r 76: two radial slots 2.2 x 3.2 through the flange at t ±4.75 over M2 inserts in Ø5.0 bosses under it (the separate 0.9 shim and the 8 x 12 board sit on the flat top, sensor up, 2.0 under the crown's code ring); LRA pad at 258.0° with the pogo-board clip beside its flex tail at 252.5° (two ribs r 76-81.7, a 0.4
7. flex slot 32 x 2.5 through the seat at 0°, offset -4; 101 vent slits 1.0 x 4.5 at z 18.5-23.0, 2.5° grid
8. motor relief at 90°

the 60 — v15 **Internal structure**

Cadrane · sheet 2 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / PA12 (production)

Process: v15: print INVERTED - the seat flange's top face on the bed, 0.20 mm, supports only under the diffuser lip (outside) and the three Ø8

mm. Tolerance ±0.1 unless stated; clearances: running 0.30,

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

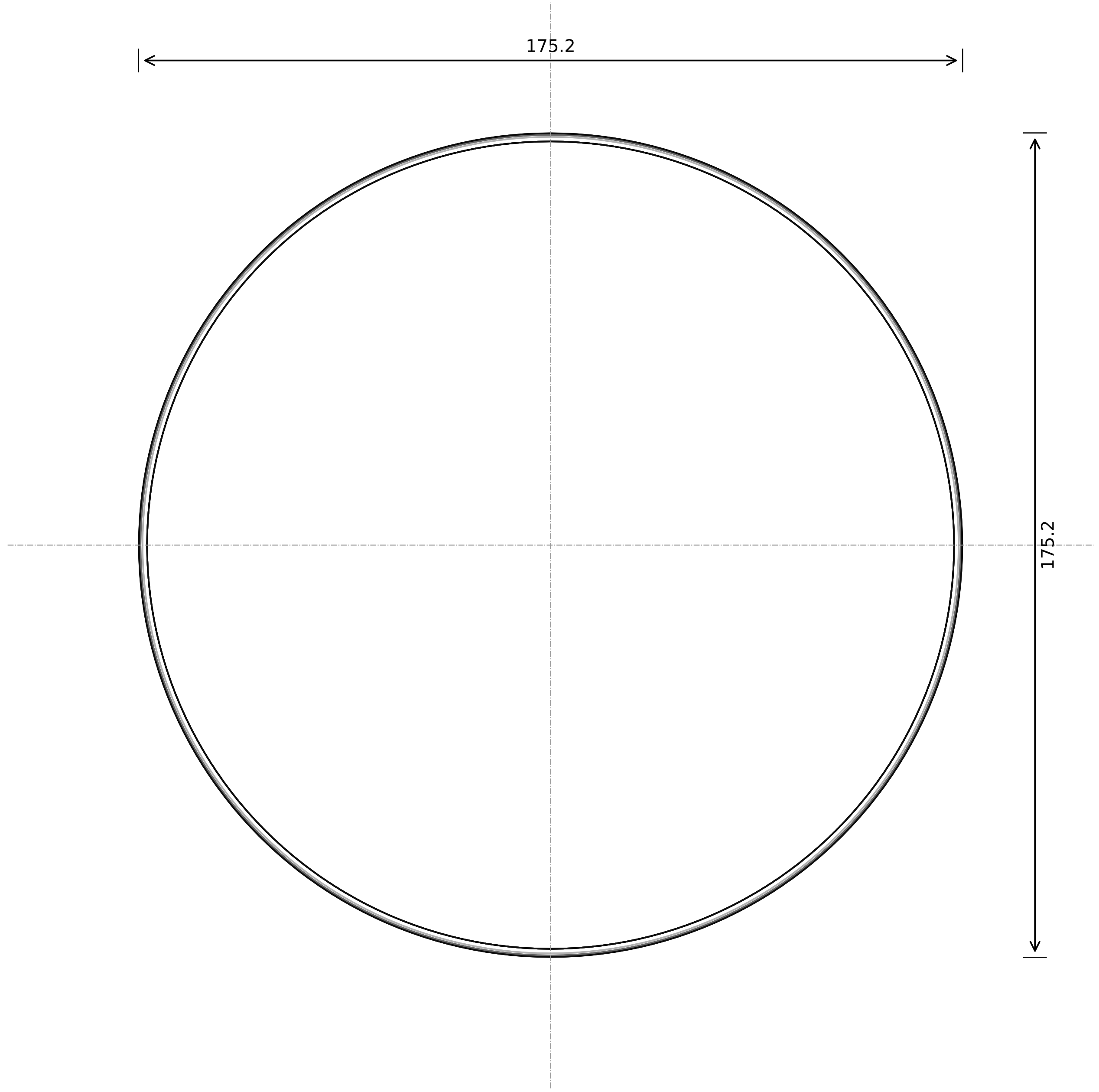
Part: internal_structure

Overall 173.4 x 173.4 x 18.0

Volume 35.73 cm³

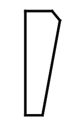
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 58 to 90, enlarged)



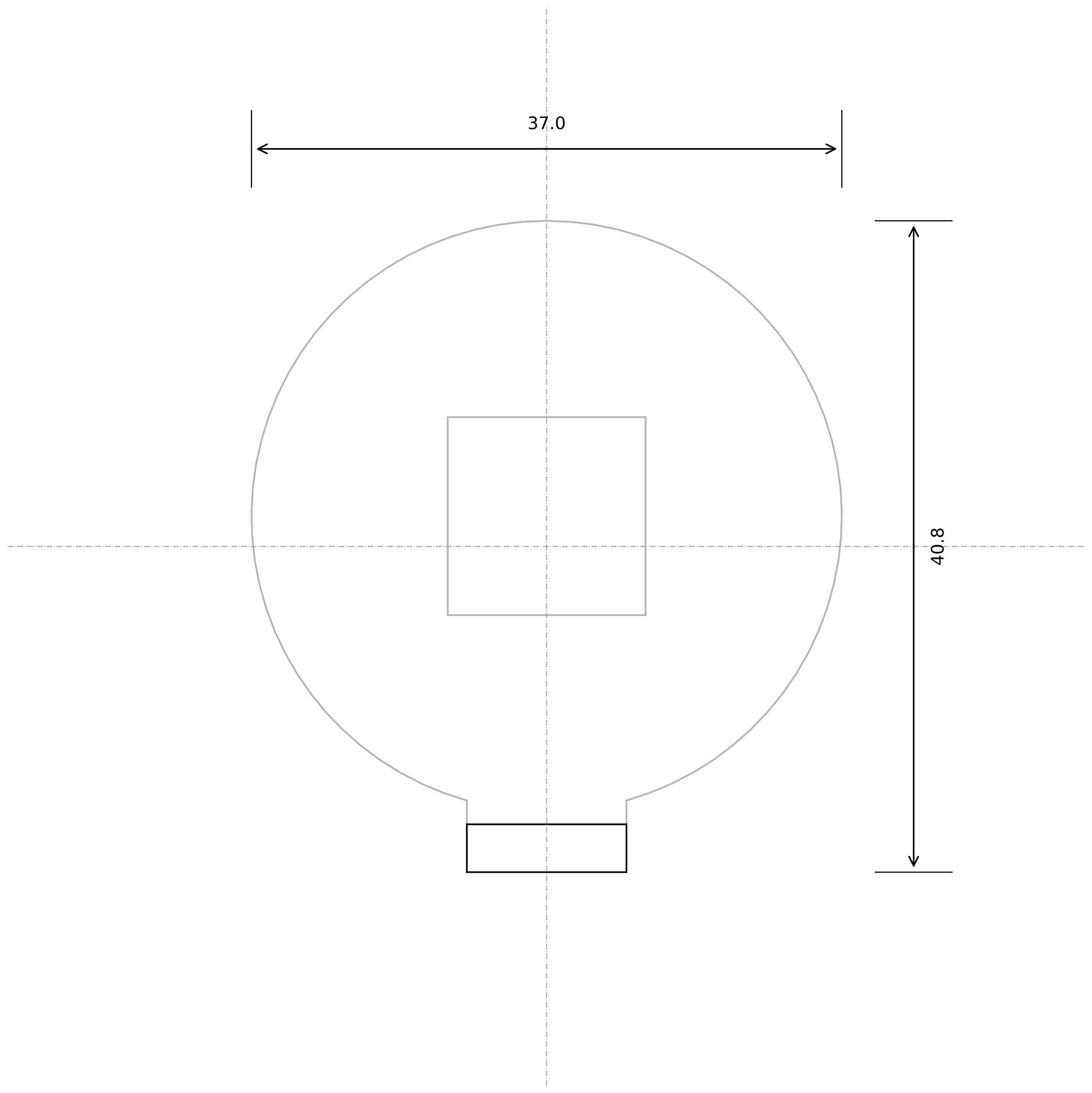
SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

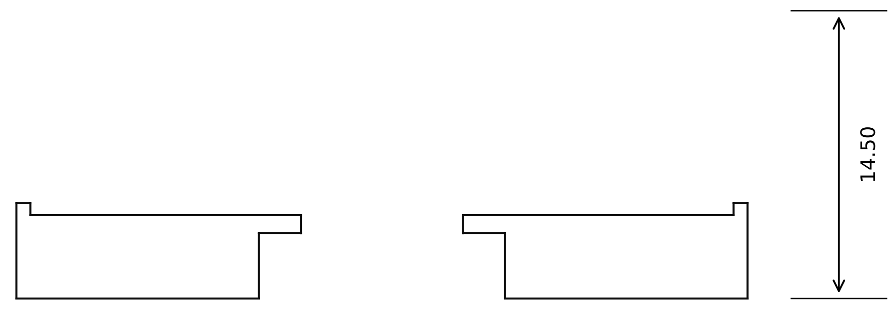
1. r 85.8-86.7 at the bottom, to r 87.7 at the top; z 8.0-13.5
2. 0.5 chamfer on the outside top edge
3. retained by the structure's lip (r to 86.7); sits on the rim ring 0.2 outside the LED strip

the 60 — v15	**Halo diffuser**
Cadrane · sheet 3 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: opal PMMA (production) / natural PETG (prototype)	**Part: halo_diffuser**
Process: print on its underside, 0.20 mm; no supports	Overall 175.2 x 175.2 x 5.5
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,	Volume 4.12 cm³
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

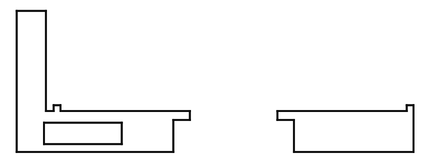


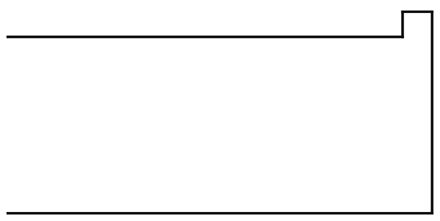
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 9 to 20, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)





Key dimensions (from the parameter block):

1. shoe Ø37 x 4.2; locating rim Ø35.6/37 x 0.6
2. commutation-board pocket 12.4 x 12.4 x 3.3; sight hole Ø9; v14: a 2.4 x 2.2 notch through the shoe's inboard wall for the sensor lead
3. push tab 3.0 x 10.0 to z 14.5 (v11: 28.5 through the deck; the servo is now beside it on the plate)
4. travel 2.4 radial in the plate's stadium hole

the 60 — v15 **Motor carriage**

Cadrane · sheet 4 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG

Process: print standing on the shoe, 0.16 mm; no supports (12 mm bridge over the sensor pocket); 4 perimeters for the tab

mm. Tolerance ±0.1 unless stated; clearances: running 0.30,

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

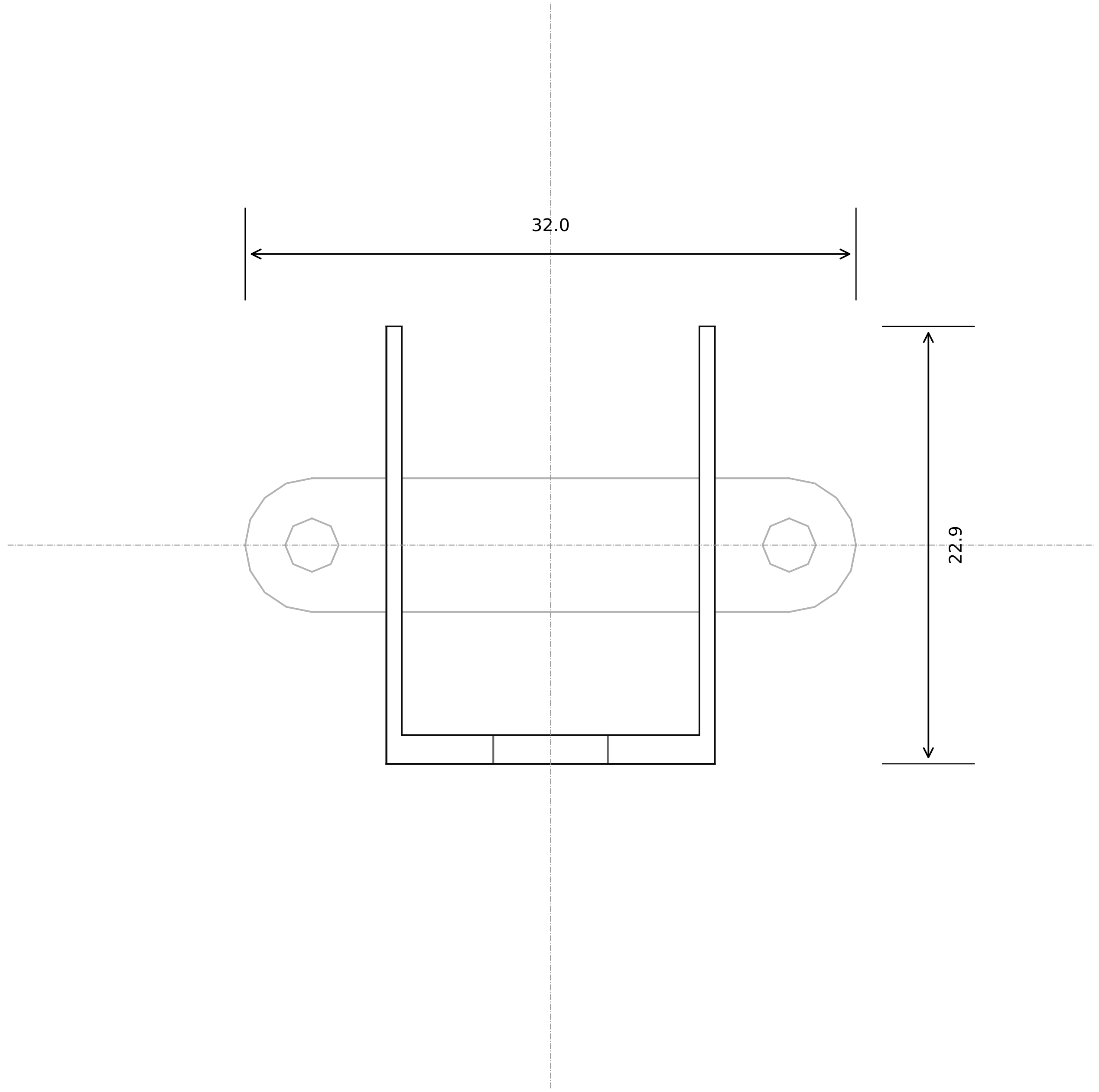
Part: carriage

Overall 37.0 x 40.8 x 14.5

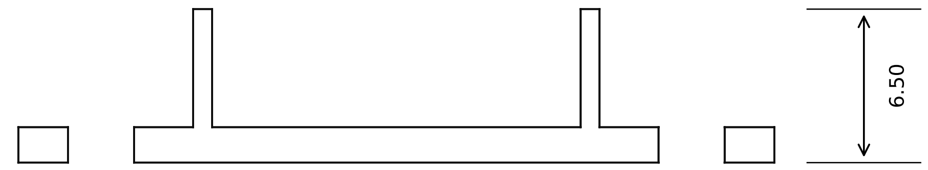
Volume 4.44 cm³

Scale: see each view (fit)

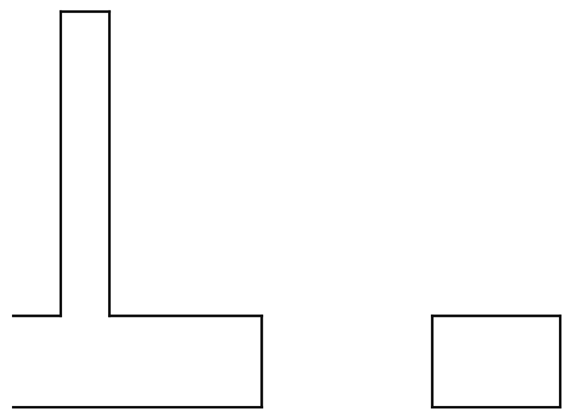
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



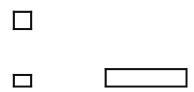
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 8 to 18, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)

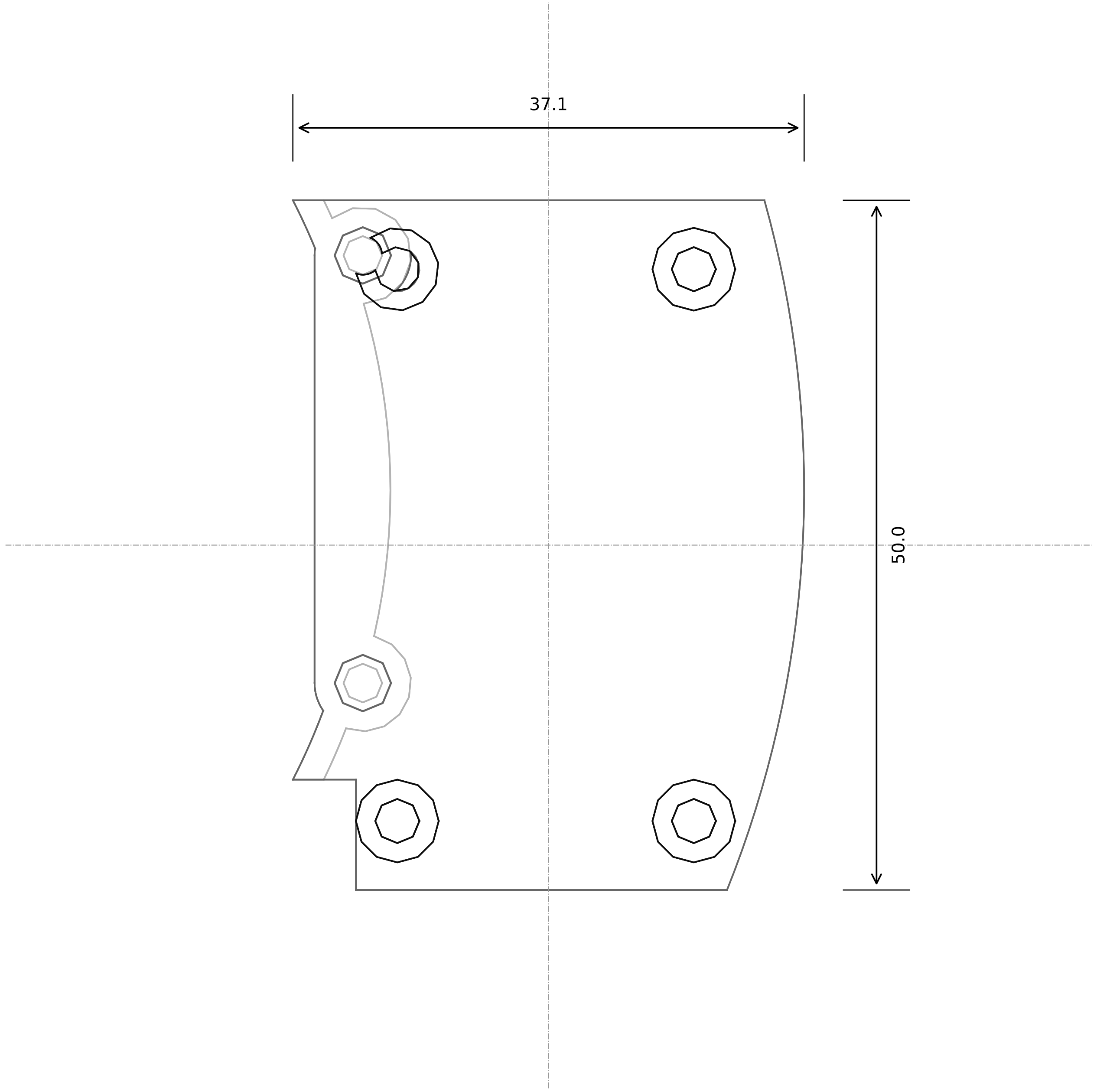


Key dimensions (from the parameter block):

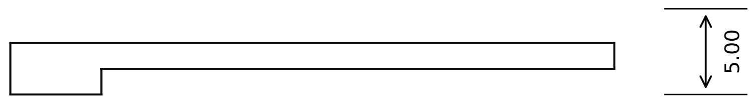
1. floorless frame round the AGFRC envelope 21.4 x 15.2 x 6.0; walls 1.0, 6.5 tall; the servo lies on the plate at r 19.0-40.4, az 90
2. open toward the carriage tab; lead notch in the inner end wall
3. 2 ears at t ±12.5, M2.5 x 4 into tapped blind holes in the plate's web

the 60 — v15	**Servo frame (on the plate)**
Cadrane · sheet 5 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: PETG	**Part: servo_mount**
Process: print flat, 0.20 mm; no supports	Overall 32.0 x 22.9 x 6.5
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,	Volume 0.64 cm³
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

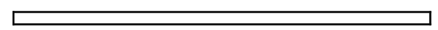


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 9 to 21, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)

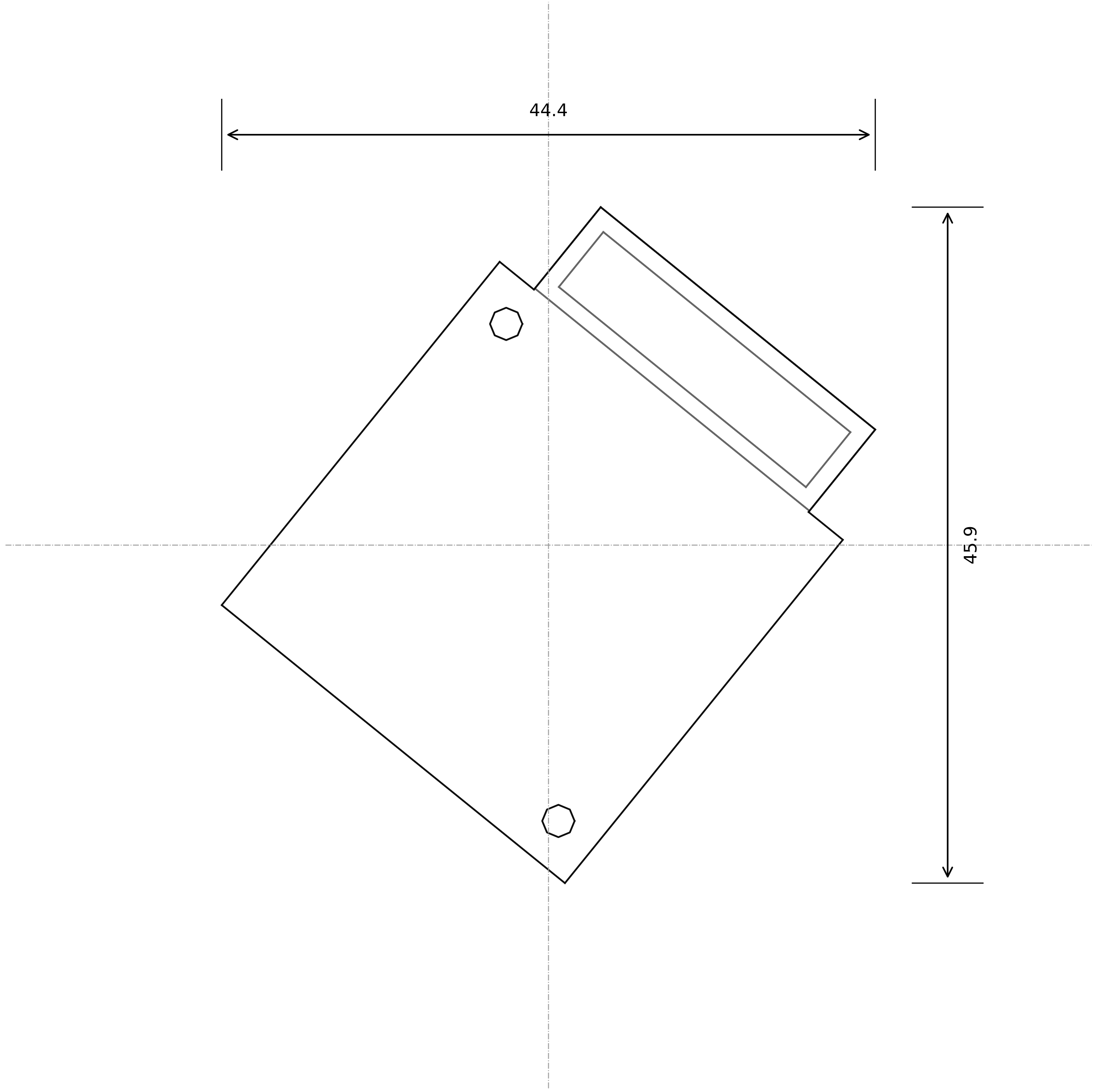


Key dimensions (from the parameter block):

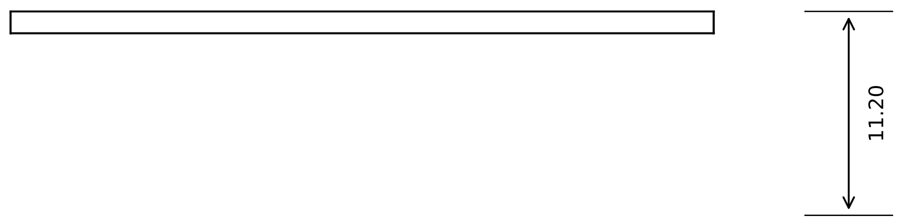
1. slab 1.5 thick at z 9.5-11.0, r 46-78, t -29..21, bridging the port slot; its outer end rests on the port face's rail (z 9.5)
2. foot strip r 46-48 on the plate; two csk ears at [(46.0, 17.0), (46.0, -14.0)], M2.5 into the web
3. four Ø6 bosses 2.0 tall with M2 inserts for the connect board (30.0 x 46.0, holes 3 in from its inner corners, 5.5 from its outer)
4. the -y inner corner cut back to x 45.5 (the audio board's USB plug is there)

the 60 — v15	**Connect-board bracket**
Cadrane · sheet 6 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: PETG Process: print standing on its straight -y edge (slab vertical, feet and bosses sideways), no supports mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	**Part: connect_bracket** Overall 37.1 x 50.0 x 5.0 Volume 2.92 cm³ Scale: see each view (fit)

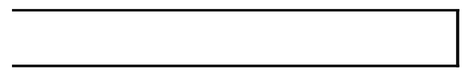
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 11 to 24, enlarged)



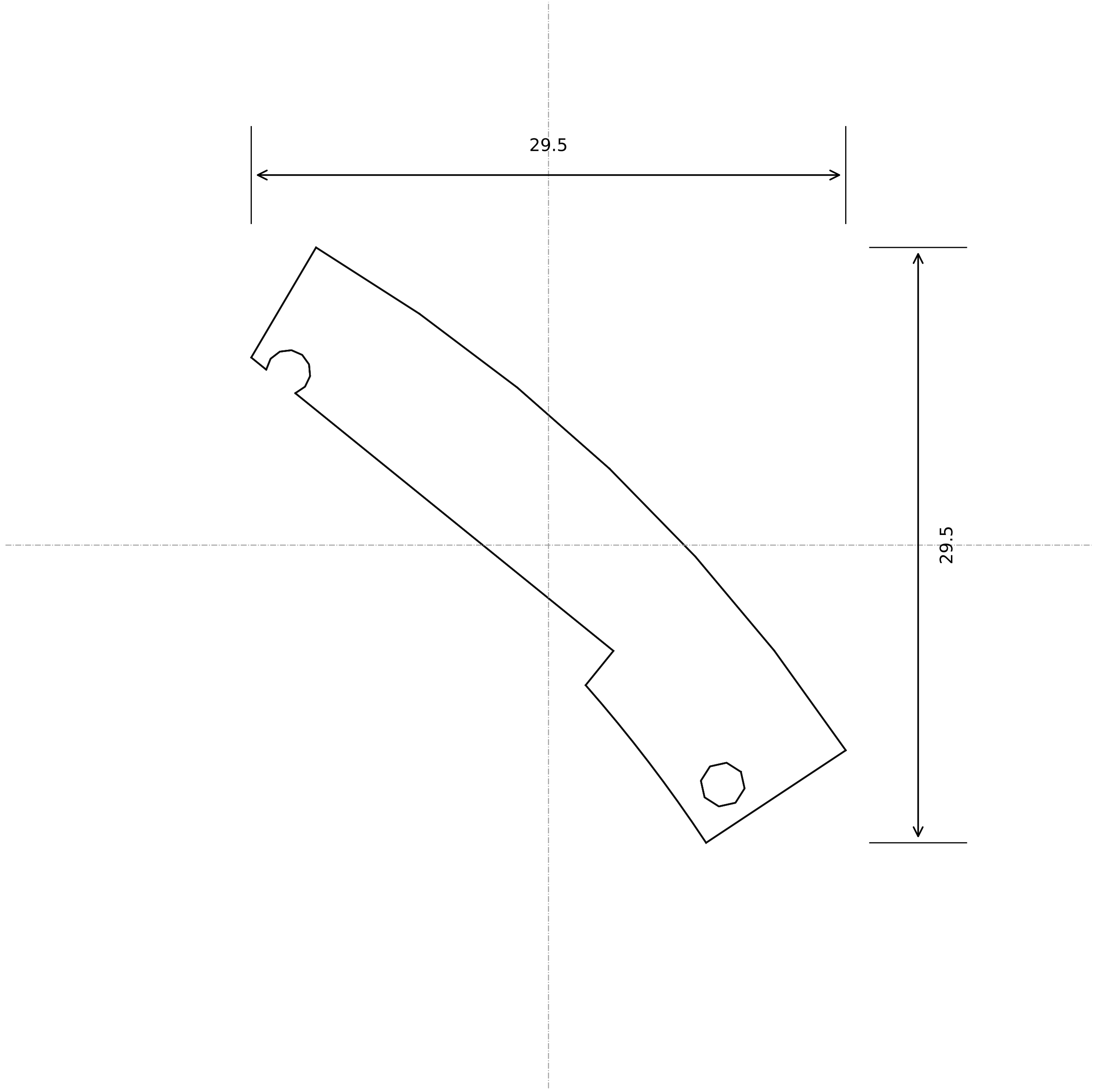
SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

1. plate 30 x 30 x 1.2 on the blower's top (z 18.3-19.5); four Ø5 x 0.5 pads for the motion board at ±12; two Ø2.2 through-holes on the blower's 24 x 24 diagonal: M2 x 18 pan heads through the board, the plate and the blower into the plate's piers
2. hood neck: 24 wide, r 66-73.2, walls 1.2, open at the bottom over the trench and toward the blower's outlet (21 x 7); on a 0.3 foam gasket on the plate top
3. blower at az 51, r 51 (Delta BFB0305HA-C 30 x 30 x 10, inlet down over the Ø22 hole in the web)

the 60 — v15	**Blower saddle + hood neck**
Cadrane · sheet 7 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: PETG Process: v15: print on its top face (the motion board's face), 0.16 mm; no supports mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	**Part: blower_saddle** Overall 44.4 x 45.9 x 11.2 Volume 1.95 cm³ Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 7 to 17, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

1. 2.5 thick at z 8.3-10.8, az 33.5-59.5, r 72.2-80.5; butts the saddle's neck (0.15 clear); two Ø2.2 at az [35.0, 58.0], r 74.5: M2 x 5 into the ring's top; on the same 0.3 gasket

the 60 — v15 | Hood lid (over the trench foot)

Cadrane · sheet 8 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG

Process: v15: print flat, 0.16 mm; no supports

mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

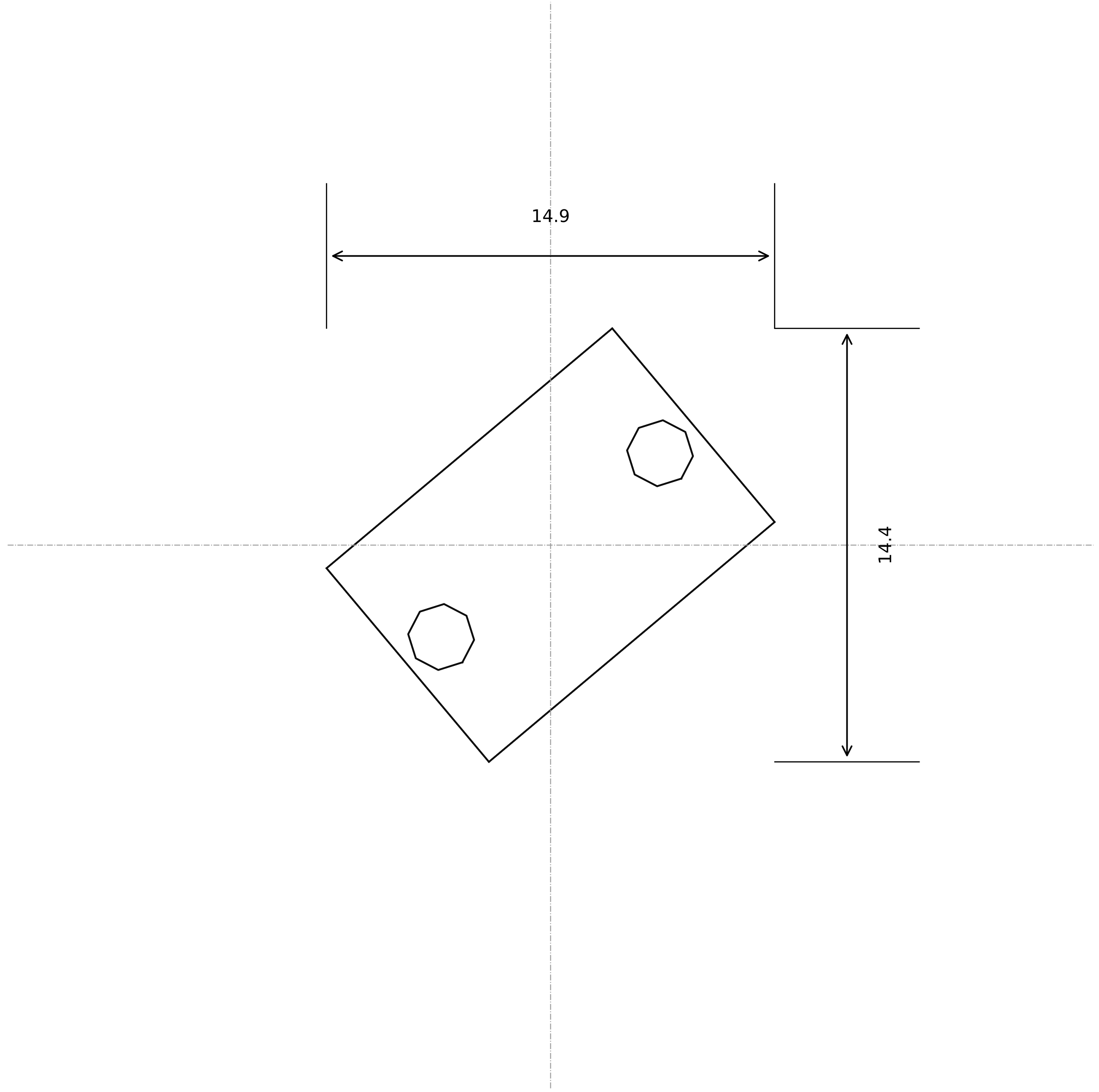
Part: hood_lid

Overall 29.5 x 29.5 x 2.5

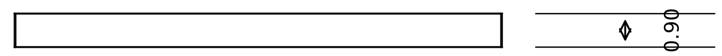
Volume 0.61 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

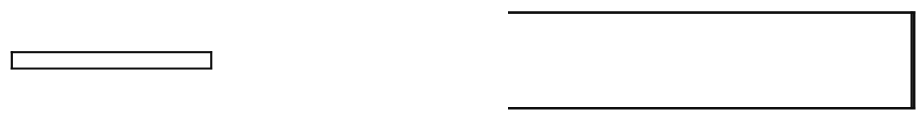


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 4 to 9, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)

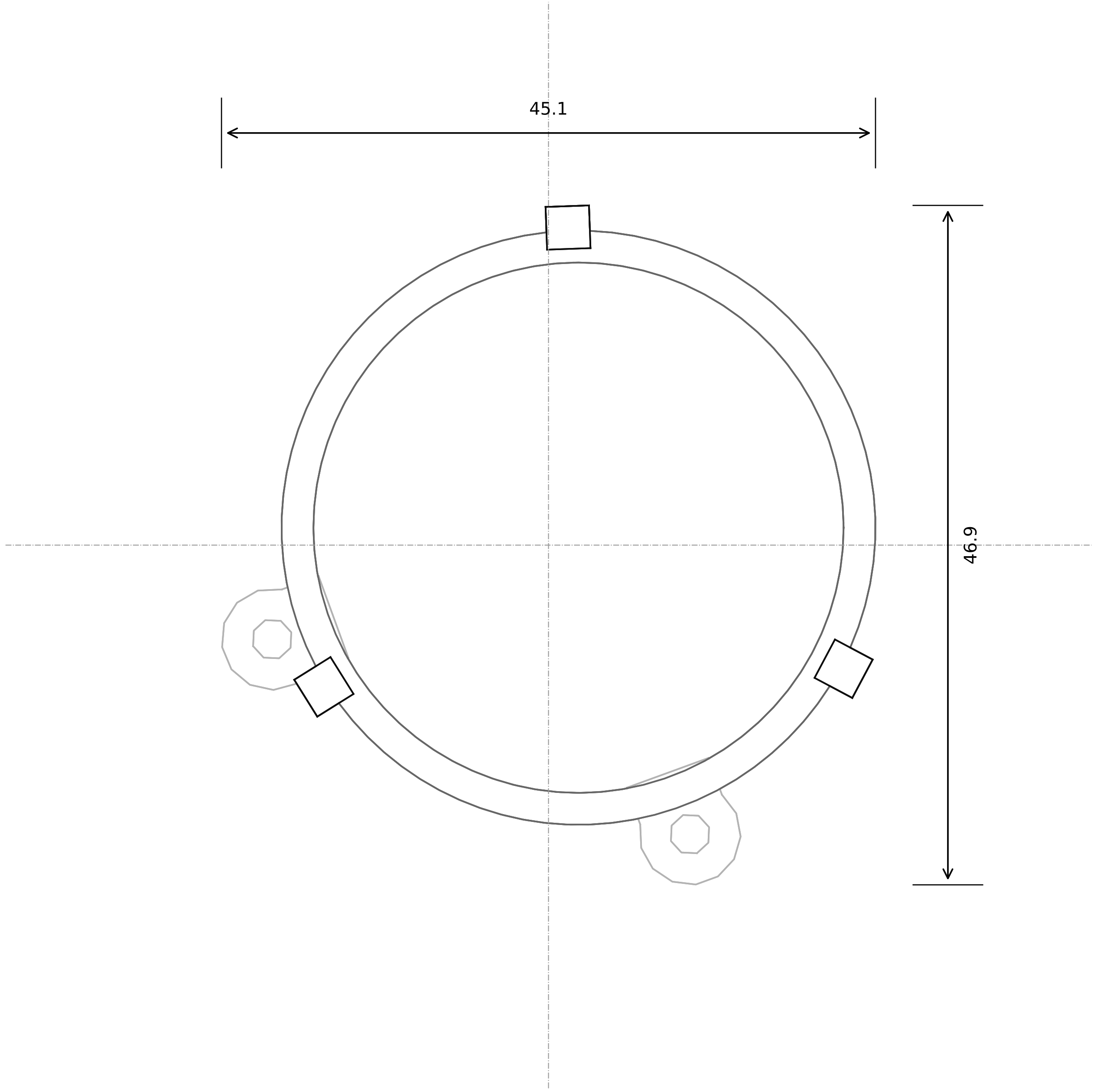


Key dimensions (from the parameter block):

1. 8.4 x 12.4 x 0.9 (nominal; family [0.7, 0.8, 0.9, 1.0, 1.1]), two Ø2.2 at ±4.75; sits on the seat flange under the encoder board, sets the 2.0 optical gap

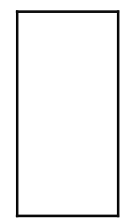
the 60 — v15	**Encoder shim (family)**
Cadrane · sheet 9 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: PETG	**Part: encoder_shim**
Process: v15: print flat, 0.10 mm; one of each height in ENC_SHIM_FAMILY	Overall 14.9 x 14.4 x 0.9
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,	Volume 0.09 cm³
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 11 to 25, enlarged)



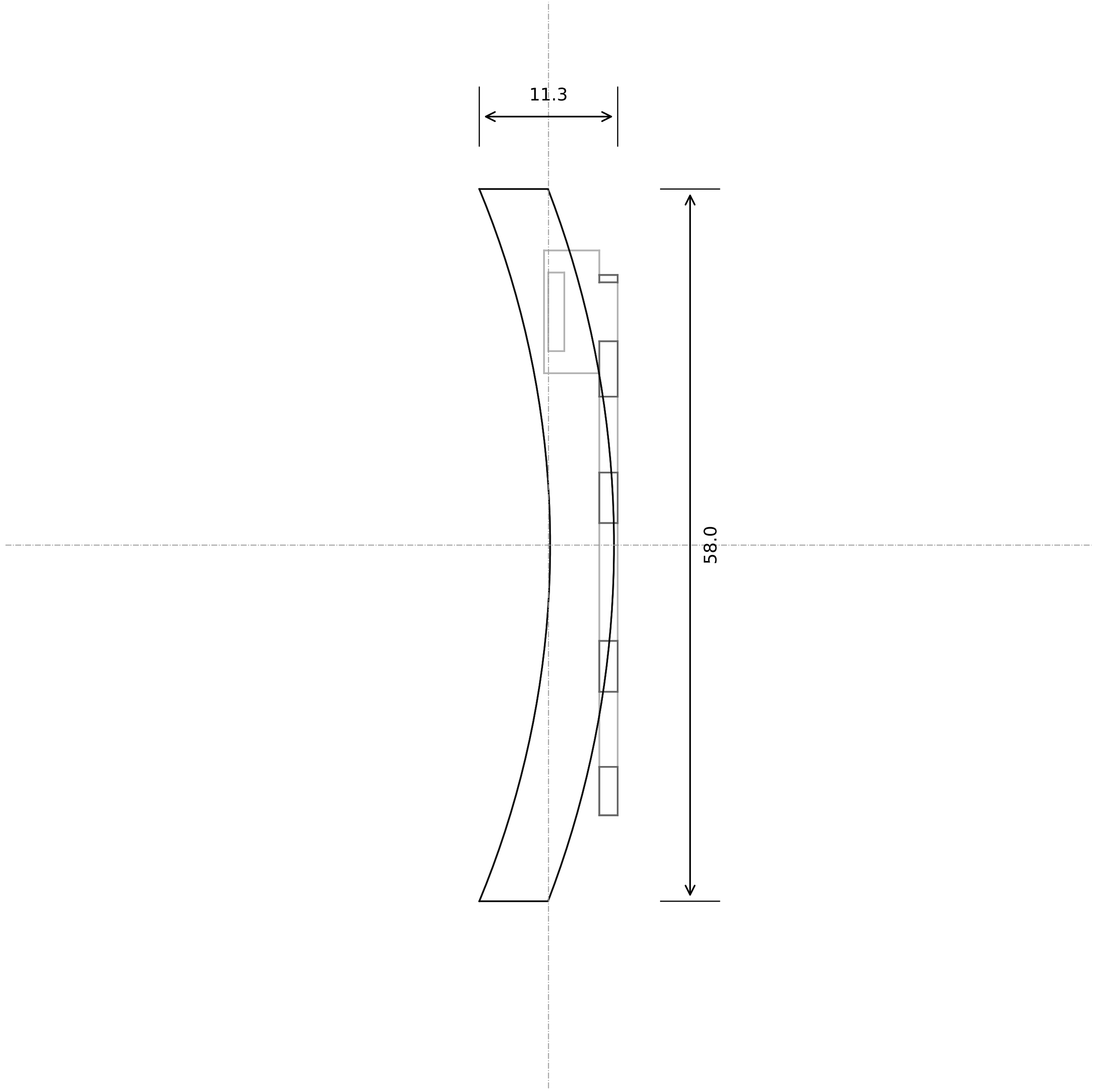
SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

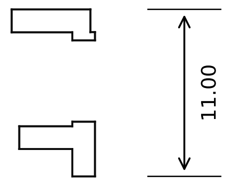
1. ring r 18.3-20.5 x 4.45 under the Ø40 flange
2. three snap fingers at [92.0, 212.0, 332.0]° from the speaker's centre (device azimuths)
3. two ears at r 22.5, [200.0, 290.0]° from the centre; M2.5 x 4 from above into the plate's web (v11: M3 from below)
4. speaker centre (-36, 47) - az 127°, r 59.2

the 60 — v15	**Speaker cradle**
Cadrane · sheet 10 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: PETG	**Part: speaker_cradle**
Process: print flat, 0.20 mm; no supports (fingers full height)	Overall 45.1 x 46.9 x 7.2
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,	Volume 1.44 cm³
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)

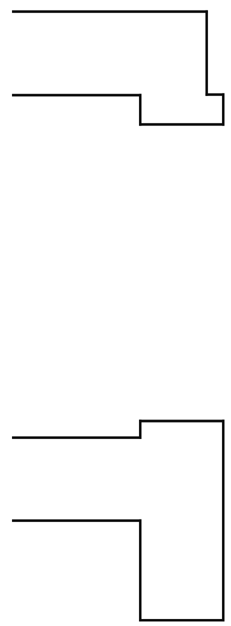
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



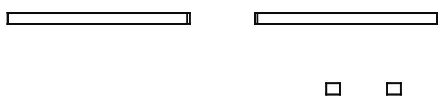
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 8, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)

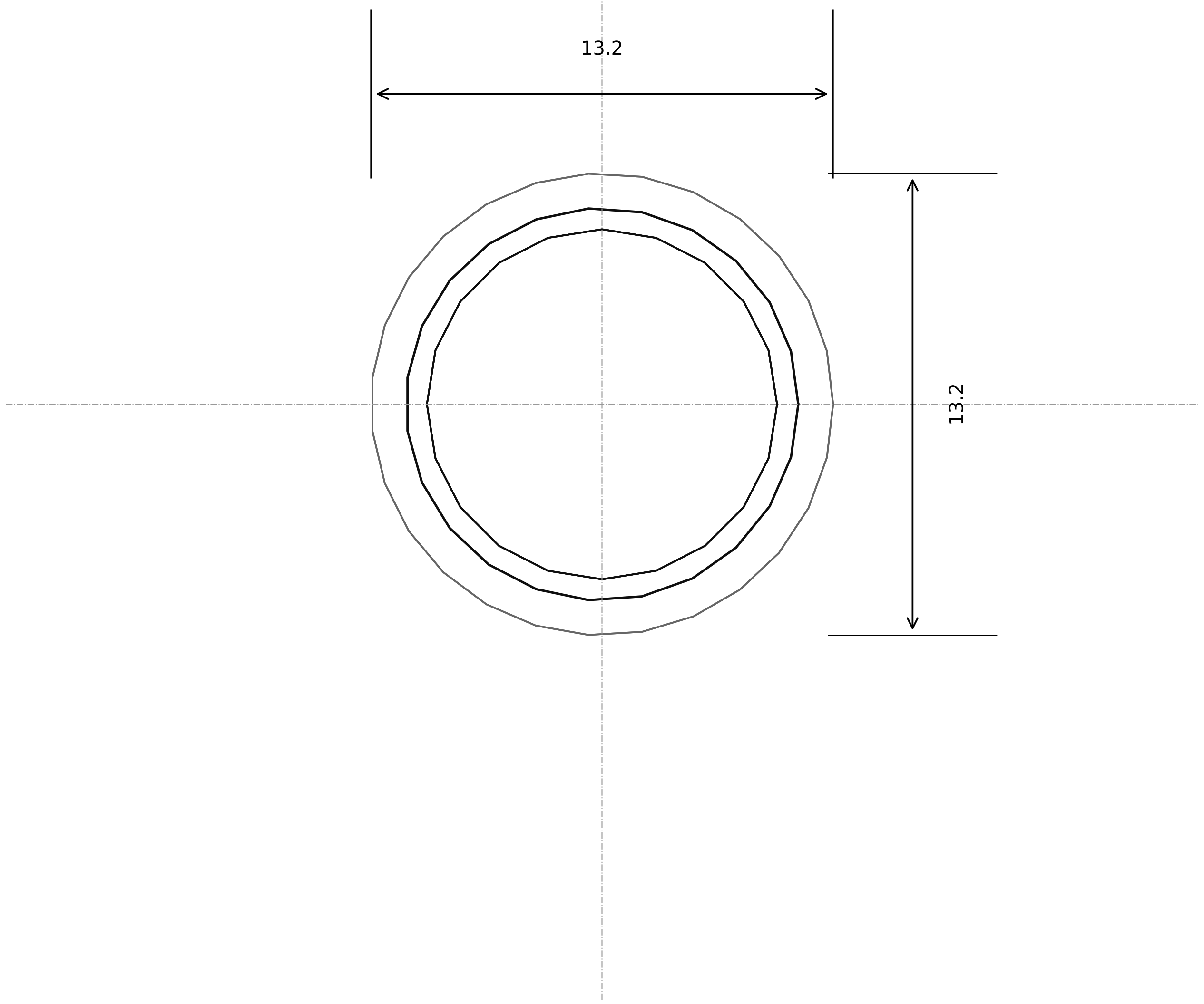


Key dimensions (from the parameter block):

1. 44.0 wide x 1.5 thick at r 79.704, z -1.5-8.0
2. barrel nose Ø6.2 at t -15.0, z 3.5; USB-C opening at t -3.0, z 4.5; jack Ø6.3 at t 9.0, z 4.6; light aperture Ø4.8 at t 19.0
3. shelf for the USB-C board; foot for the light-sensor board; rail along the top on the plate (1.5 tall, 58 wide, inner edge r 75.7, flat front in the face's plane, corners cut 45° to stay inside the diffuser) with two Ø2.2 holes at r 78.2, t ±26: M2 screws into the plate

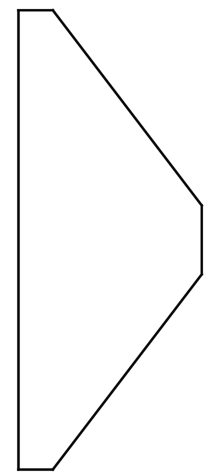
the 60 — v15	**Rear port face**
Cadrane · sheet 11 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: PETG Process: print lying on its outer face, 0.16 mm; no supports (the socket holes print vertical) mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	**Part: port_face** Overall 11.3 x 58.0 x 11.0 Volume 1.01 cm³ Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 3 to 9, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

1. Ø13.2 x 4.0, 90° V 1.3 deep with a 0.6 flat; bore Ø10 press over a 623ZZ

the 60 — v15 Wheel V-collar (x3)

Cadrane · sheet 12 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / POM (production)

Process: print axis vertical, 0.12 mm

mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: collar_0

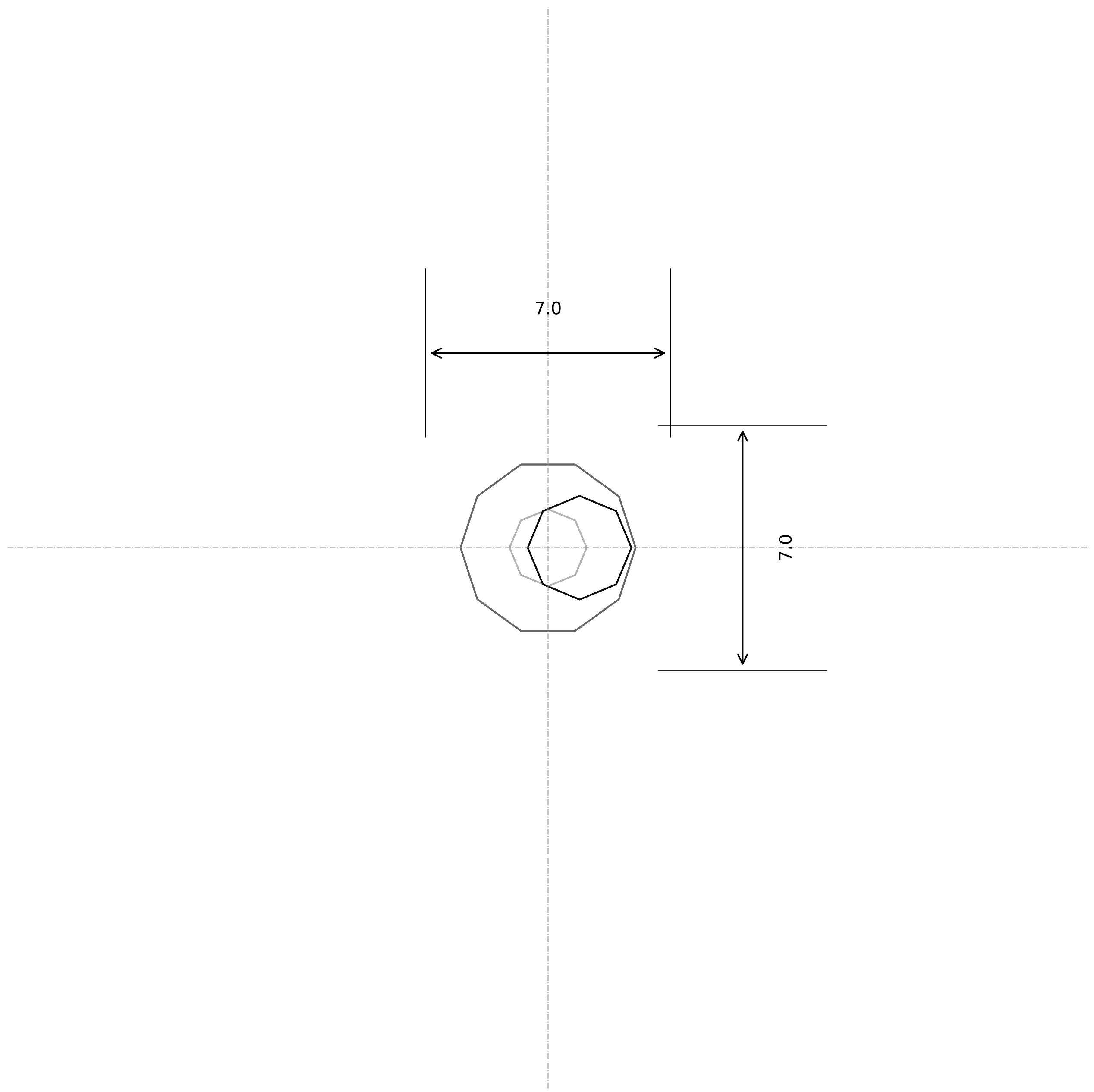
Overall 13.2 x 13.2 x 4.0

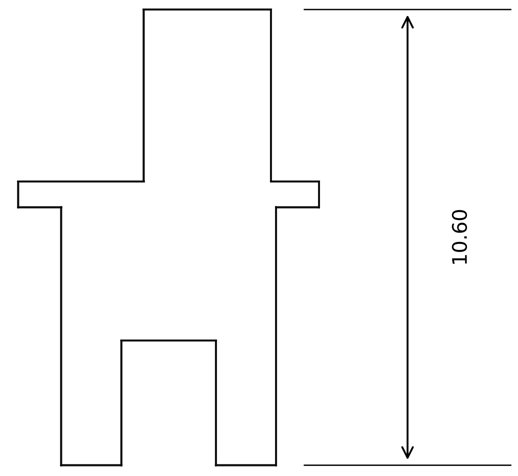
Volume 0.15 cm³

Scale: see each view (fit)

SECTION A-A (front, plane y = 0; x across, z up)

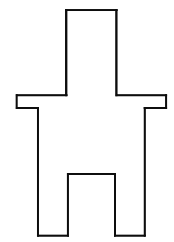
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

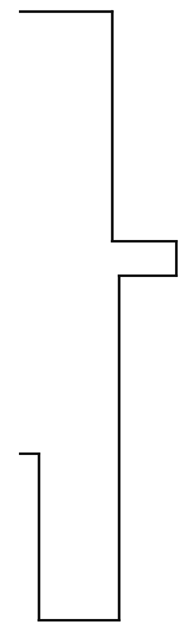




DETAIL C (section A-A, x 2 to 6, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



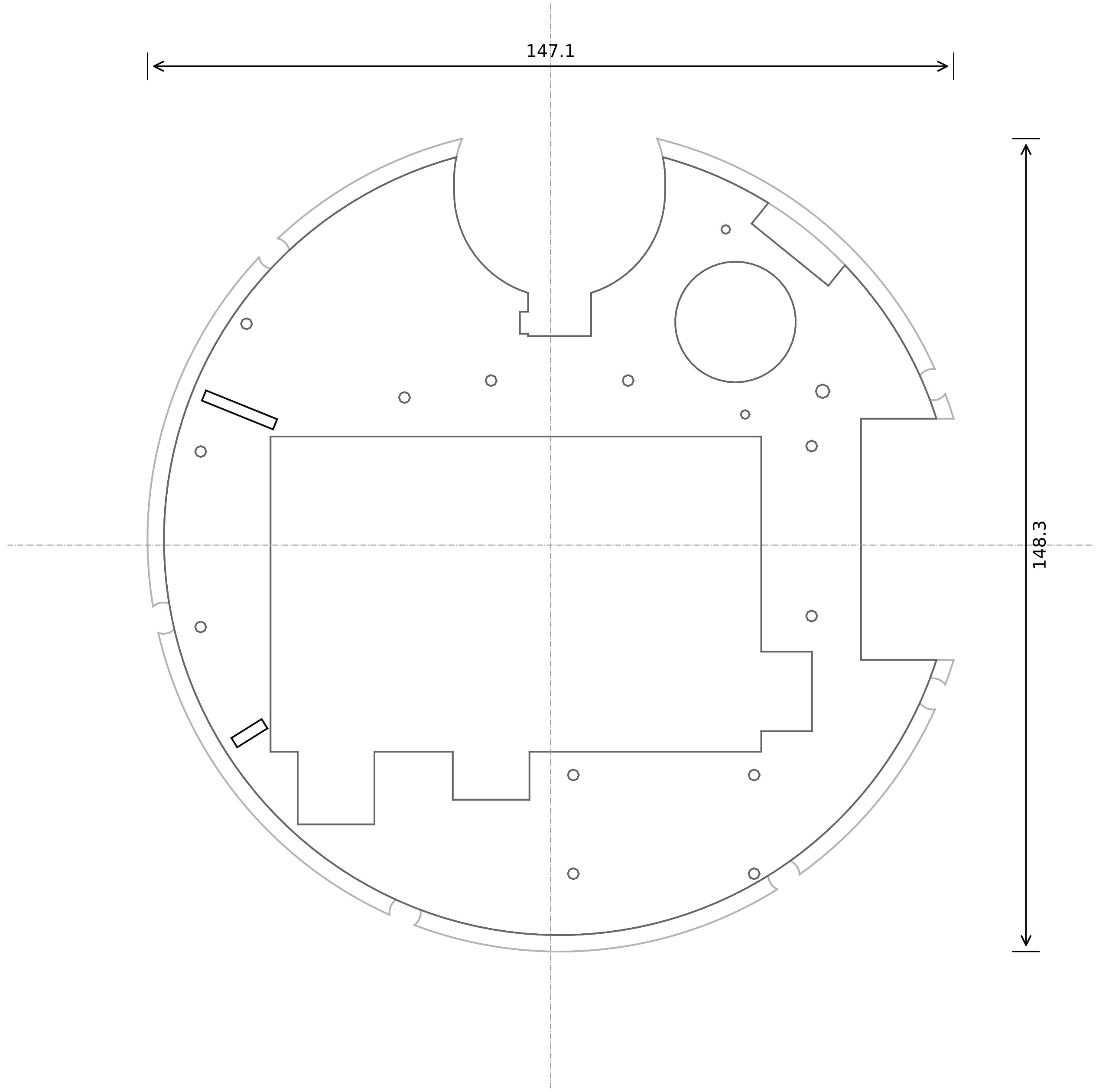


Key dimensions (from the parameter block):

1. Ø5.0 shank x 6, Ø7.0 x 0.6 flange, Ø3.0 pin x 4.0 offset 0.9; 2 mm hex socket

the 60 — v15	**Eccentric bush (x3)**
Cadrane · sheet 13 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: PETG (prototype) / brass (production) Process: print stem down / pin up, 0.12 mm; the 1 mm flange ledge is the only overhang mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	**Part: bush_0** Overall 7.0 x 7.0 x 10.6 Volume 0.16 cm³ Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 44 to 76, enlarged)

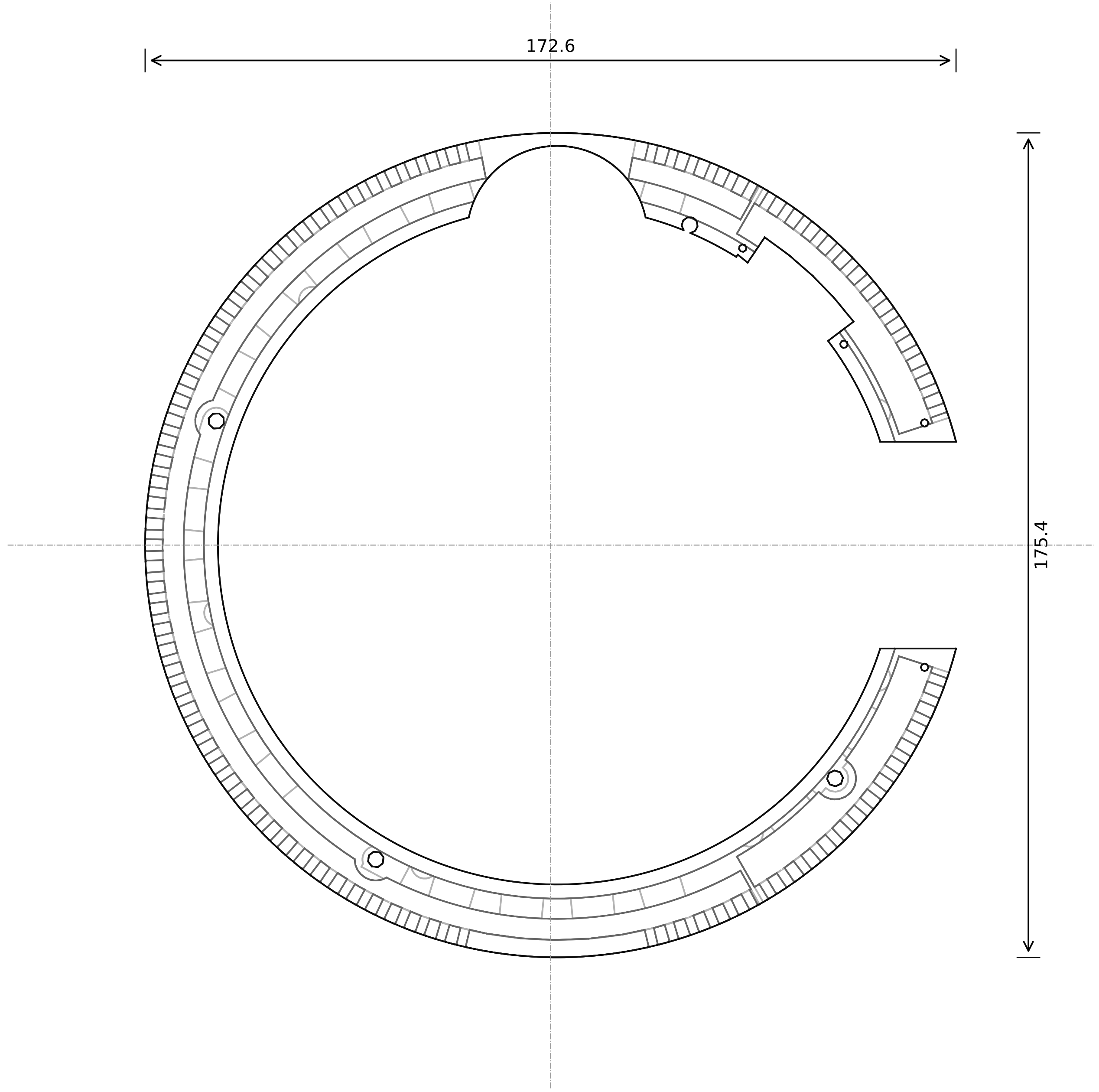
SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

1. r < 75.2 x 8.0: 2 web (z 6-8), 5 duct (z 1-6) closed by a 1 mm plate in a recess on a 0.2 gasket in a groove; Ø7 piers 2.5 into the duct under every screw from the top; rebate r 72.2-75.2, z 5-8 for the ring's flange on a 5 shoulder, 7 x M3 csk from below at r 73.7; the ring's bond screw M3 tapped 3.0 into the shoulder at az 67.
2. duct: collector r 66-70.2 (front and sides), 19 straight channels 3 x 5 on a 4 pitch beside the Pi window, rear plenums r 55+; 2.5 walls round every through-cut; intake passages as closed tunnels z 1-4.5 through the duct wall and shoulder (17 x 9 wide) + 3 exhaust tunnels 10 wide at az [311.0, 320.0, 329.0]
3. BLOWER (v15, fitted): Ø22 inlet hole through the web at az 51, r 51; discharge trench 18 wide from r 66.8 out through the duct wall and the shoulder, widening to az 37-56 (open-topped, the hood seals it); the 2 mm wall at r 64.8-66.8 keeps it off the suction side
4. Pi window (Pi at (-52, -38), with the USB-C plug head, HDMI plug and USB-A plug notches), carriage hole, port slot; 12 x M2.5 tapped blind 3.5 deep from the top (converter, audio, servo frame, connect bracket, speaker cradle) + 2 x M2 (the blower) + THE chassis bond (M3) at (48.0, 27.0), each on a pier; 9 x M2.5 tapped fro
5. 2 ribs 2.0 x 6.0 tall on the top face at r 56-70, az [158, 212]

the 60 — v15 Cadrane · sheet 14 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	**Base plate - aluminium core (the heatsink)**
Material: 6082 aluminium, CNC two-sided; EXTERNAL faces black hard anodised; INTERNAL faces chromate or bare - no masked pads (design Process: machined, not printed (print a PETG stand-in without the duct for the fit check). mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	**Part: base plate** Overall 147.1 x 148.3 x 14.0 Volume 40.76 cm³ Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 56 to 88, enlarged)

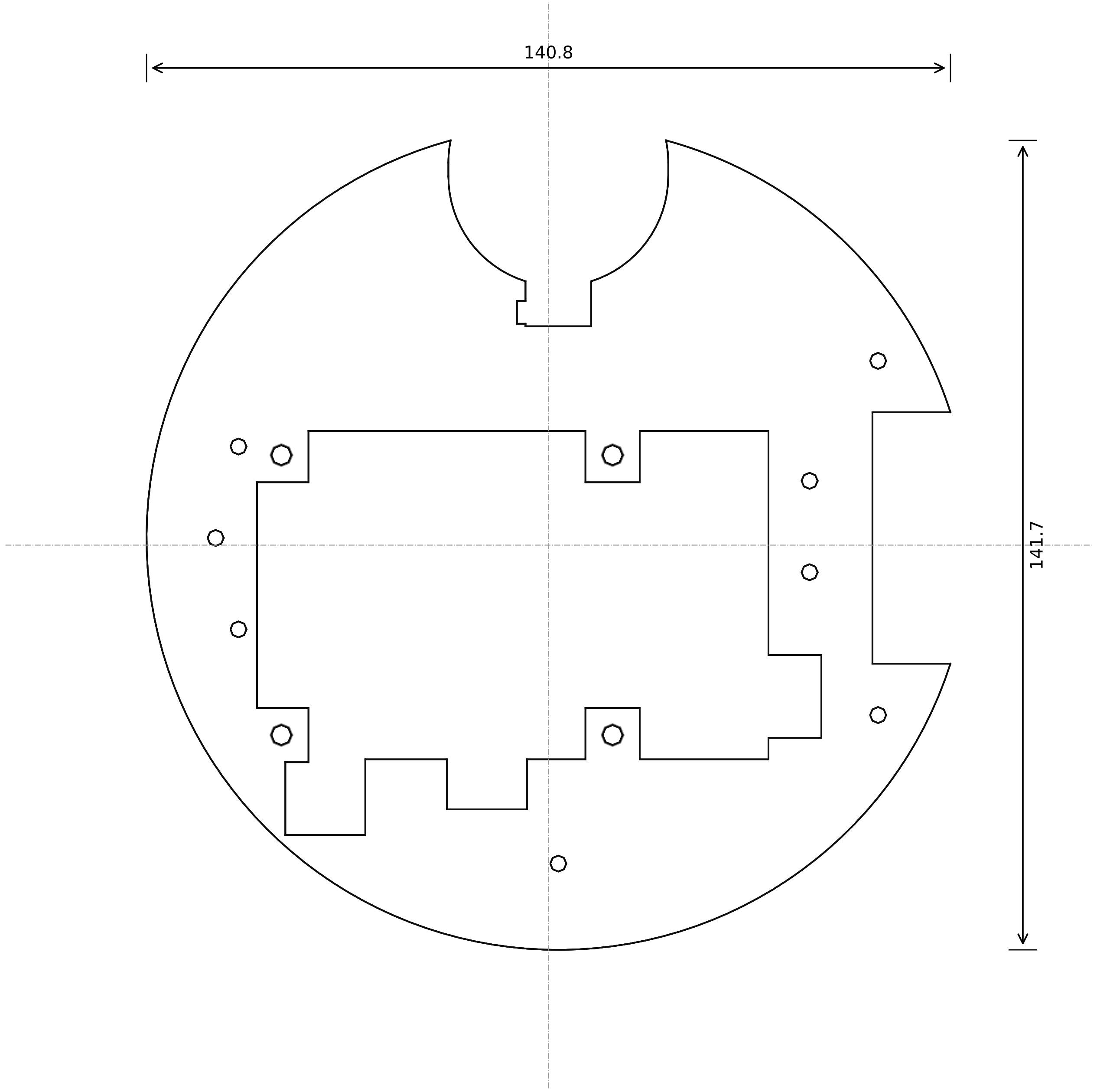
SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

1. r 75.2-87.7 x 8.0; 92 obround openings 2.0 x 4.5 (z 1.55-6.05) on 3° through the 3.7 outer wall, az 1.5 + 3k, mirrored about 0-180, none across the port face (342-18): 64 intake (60-300) + 28 exhaust (18-60, 300-342); 0.3 x 45° polished at every mouth; the upper land plain (no reeding - Ryan, 7 Sep); 0.3 chamfers on both c
2. underside groove behind the openings: r 79.5-84.0 (intake arc) / r 76.5-84.0 (exhaust arcs), z 0-6.55, walls 1.2° at 60 and 300; the pad closes it; the 0.8 undercut under the outer wall (r 84.0-87.7) opens into it all round both arcs; the blower's trench: flange notched and inner land cut away az 37-56 to r 79.0, the groove's roo
3. inward flange r 72.2-75.2, z 5-8, 7 x M3 tapped blind from below at r 73.7, az [22.5, 78.75, 135, 191.25, 247.5, 303.75, 337.5]; the dedicated BOND screw's Ø3.4 clearance at az 67.5, r 73.7 (M3 from the top into the core, star washer under the head, both faces bare)
4. passages across the inner land (r 75.2 to the groove), 3.5 deep, the pad closes them: 17 intake 9 wide (536 mm2) on the 11.25° grid, 3 exhaust 10 wide at az [311.0, 320.0, 329.0] (the -y circuit; the +y circuit exhausts through the trench)
5. 3 x M3 csk from below at r 77.2, az [160, 240, 320] (the structure's pillars); 2 x Ø1.6 tapped M2 at r 78.2, t ±26 (the port face's rail); port slot 44.0 wide

the 60 — v15	**Rim ring (stainless)**
Cadrane · sheet 15 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
Material: stainless steel 304/316, machined; BARE: electropolish (the 0.3 chamfers at every opening mouth and both edges), then brush the	**Part: rim_ring_STEEL**
Process: machined, not printed	Overall 172.6 x 175.4 x 8.0
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,	Volume 26.17 cm³
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)

DETAIL C (section A-A, x 40 to 72, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)

Key dimensions (from the parameter block):

1. disc r 72.1 x 1.0, flush in the core's recess on a 0.2 die-cut gasket; same Pi window, carriage hole and port slot as the core
2. four 9.5 square lands reaching into the Pi window at the Pi's holes [(-48.5, -34.5), (9.5, -34.5), (-48.5, 14.5), (9.5, 14.5)], Ø2.7 csk from below: the Pi and adapter stack screws to them (M2.5 x 6 into the standoffs)
3. 9 x M2.5 countersunk; unscrew to clean the fin channels

the 60 — v15   Closing plate (aluminium)

Cadrane · sheet 16 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: 1 mm aluminium, laser-cut, black anodised (outer face)	**Part: closing_plate_AL**
Process: laser-cut, not printed	Overall 140.8 x 141.7 x 1.0
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,	Volume 9.30 cm³
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)